

Involvement of Nitric Oxide and N-Methyl-D-Aspartate in Acute Hypoxic Altitude Convulsion in Mice

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Background: Altitude convulsion is a rather specific form of experimental convulsion which is induced by acute exposure to a hypobaric hypoxic condition. Several neurotransmitters have been shown to be involved in the mechanisms of altitude convulsions. However, their roles and interaction were not clear. **Hypothesis:** The novel neurotransmitter nitric oxide (NO) may be involved in the mechanisms of altitude convulsion through its neuronal signalling roles in relation to the NMDA receptor. **Methods:** There were 177 mice intraperitoneally administrated (i.p.) with several drugs. The altitude convulsion threshold (ACT) was used as an index to evaluate the acute hypoxic tolerance. **Results:** NO synthesis precursor, L-arginine (20, 40, 200, 800 mg/kg), resulted in a dose-dependent decrease in the ACT in mice, while the NO synthase (NOS) inhibitor, N^G-nitro-L-arginine-methyl ester (L-NAME, 1.25, 2.50, 5.00 mg/kg, i.p.) increased the ACT. Pretreatment with L-NAME (5.0 mg/kg) prior to L-arginine (200 mg/kg) administration prevented the effect of decreasing ACT caused by L-arginine. Similarly, N-methyl-D-aspartate (NMDA, 2.5, 10.0, 20.0 mg/kg, i.p.) yielded a decrease in the ACT in mice and this decrease in hypoxic tolerance caused by NMDA can be prevented by pretreatment with either NMDA receptor antagonist 2-amino-5-phosphovalerate (AP-5, 20.0 mg/kg, i.p.) or NOS inhibitor L-NAME (5.0 mg/kg, i.p.). **Conclusions:** These findings suggest an important signalling role for nitric oxide and NMDA in the development of altitude convulsion and further support the hypothesized relationship between NMDA-receptor mediated neurotoxicity and nitric oxide.

EXPOSURE TO ACUTE hypobaric hypoxia in several species of animal leads to seizures and mortality. The phenomenon was termed as altitude convulsion and it represents the critical phase of the body response to a sudden ascent of altitude level. The severity of the altitude convulsion depends on the rate of ascent and it is assessed by the value of altitude convulsion threshold, ACT which is defined as the time required to initiate the convulsion (12). Therefore, the ACT value is an important index in studying the mechanisms of altitude convulsion as well as acute hypoxic tolerance.

Nitric oxide (NO), formerly known as endothelium-derived relaxing factor (EDRF, 5) is formed when the amino acid L-arginine is enzymatically oxidized by nitric oxide synthase (NOS, 1). The discovery of NO as neurotransmitter has radically altered thinking about synaptic transmission because NO is a labile free radical and it is

not stored in synaptic vesicles as other neurotransmitters. Instead, it must be synthesized on demand from its precursor L-arginine by NOS. The observation that the excitatory neurotransmitter glutamate acting at N-methyl-D-aspartate (NMDA) receptors stimulates the formation of NO provides definite evidence for a role of NO in synaptic transmission (6). This NMDA-induced stimulation of NO formation is blocked by NOS inhibitors which also block the NMDA stimulation of cGMP in the brain slices (1).

Several NMDA receptor antagonists possess well-known anticonvulsant effects in different animal models of seizures (13) while L-arginine has been proved to possess proconvulsant effects through the formation of nitric oxide (14). However, the combined effects of NMDA and NO on convulsion, as well as their roles in the mechanisms of hypobaric hypoxic convulsion (altitude convulsion), are still to be determined. In this study we investigated the effects of NMDA and NO transmissions on the acute hypoxic tolerance of altitude convulsion in mice.

METHODS

Male ICR mice weighing between 18 and 22 grams were used in all groups of this study. The mice were housed for at least 5 d before the experiments. They were kept in groups of five per cage and had free access to food and water at the room temperature within 22-24°C at all time. On the day of experiment, the mice were placed in a decompression chamber and subjected to severe hypobaric hypoxia at the rate of ascent of 14,000 ft·min⁻¹, (i.e., 35,000 ft within 150 s). The ACT in seconds (i.e., the time required for initiating the convulsion)

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TABLE I. EFFECTS OF L-ARGININE OR L-NAME ADMINISTRATION ON THE ALTITUDE CONVULSION THRESHOLD (ACT) IN MICE.

Group	Altitude Convulsion Threshold (s)
I.	
Saline	227.8 ± 8.6
L-arginine 20 mg · kg ⁻¹	184.2 ± 17.0*
L-arginine 40 mg · kg ⁻¹	173.5 ± 15.9*
L-arginine 200 mg · kg ⁻¹	167.8 ± 19.2*
L-arginine 800 mg · kg ⁻¹	161.2 ± 9.6**
II.	
Saline	210.8 ± 8.0
L-NAME 1.25 mg · kg ⁻¹	247.8 ± 26.1
L-NAME 2.50 mg · kg ⁻¹	251.5 ± 36.0
L-NAME 5.00 mg · kg ⁻¹	268.8 ± 19.3*
III.	
Saline	221.3 ± 6.5
L-arginine 200 mg · kg ⁻¹	160.0 ± 15.6**
L-arginine 200 mg · kg ⁻¹ + pretreatment with L-NAME 5.0 mg · kg ⁻¹	202.5 ± 8.1†

Data are expressed as the mean ± S.E.M. of 6 mice.

* Significantly different from the values in corresponding saline-treated mice, $p < 0.05$; ** $p < 0.01$.

† Significantly different from the values in L-arginine (200 mg · kg⁻¹)-treated mice, $p < 0.05$.

was used as an index for the assessment of hypoxic tolerance. The time was rated by an observer who was blind to the drugs tested in this study. Details of the methods were described previously (12).

A total of 177 mice were studied in three sets of experiments. The first group was given either single intraperitoneal injection of 0.5 ml saline solution of the NO precursor L-arginine (20, 40, 200, 800 mg · kg⁻¹) or an NO synthase inhibitor N^G-nitro-L-arginine-methyl ester (L-NAME, 1.25, 2.50, 5.00 mg · kg⁻¹) 15 min before they were placed in a decompression chamber. To investigate whether pretreatment with L-NAME alters the effects induced by L-arginine, L-NAME (5.0 mg · kg⁻¹, i.p.) was given 5 min prior the administration of L-arginine (200 mg · kg⁻¹, i.p.). Similarly, the second group was given either single intraperitoneal injection of NMDA (2.5, 10.0, 20.0, mg · kg⁻¹) or an NMDA-receptor antagonist 2-amino-5-phosphovaleate (AP-5, 10.0, 20.0, 40.0 mg · kg⁻¹) or vehicle (normal saline as control group) 15 min before they were placed in a decompression chamber. To investigate whether pretreatment with AP-5 alters the effects induced by NMDA, AP-5 (20.0 mg · kg⁻¹, i.p.) was given 5 min prior the administration of NMDA (10 mg · kg⁻¹, i.p.). The third group tested the combined effects of NMDA, AP-5, and L-NAME on hypoxic tolerance. Either AP-5 (20 mg · kg⁻¹ i.p.) or L-NAME (5 mg · kg⁻¹, i.p.) was administered 5 min prior NMDA (10 mg · kg⁻¹, i.p.) injection. All the drugs used in this study were purchased from Sigma Chemical Co., St. Louis, MO.

Differences between control and drug-treated animals were analyzed using Student's two-tailed *t*-test for unpaired observations. A *p*-value of <0.05 was considered significant. Data are presented as mean ± standard error of the mean (S.E.M.).

RESULTS

A dose-related decrease in the altitude convulsion threshold (ACT) in second, was found after administra-

tion of L-arginine (20, 40, 200, 800 mg · kg⁻¹, Table I). The decrease in ACT (i.e., decrease in hypoxic tolerance) was significantly different from control group at all the L-arginine doses which we tested. As to the effects of NOS inhibitor L-NAME, it slightly increased the ACT at the dose levels of 1.25 and 2.50 mg · kg⁻¹, and the increase reached a statistically significant difference when the dose of L-NAME went up to 5.00 mg · kg⁻¹ (Table I). Pretreatment with L-NAME (5.0 mg · kg⁻¹) was able to reverse the effect of L-arginine on decreasing ACT in that ACT value was reversed from 160.0 ± 15.6 s (L-arginine 200 mg · kg⁻¹) to 202.5 ± 8.1 s (pretreatment with L-NAME 5.0 mg · kg⁻¹) (Table I).

In the group of animals treated with either NMDA or NMDA receptor antagonist AP5, a decrease in the time required for initiating altitude convulsion was noted after administration of NMDA (2.5, 10.0, 20.0 mg · kg⁻¹, Table II). NMDA receptor antagonist AP5 itself (10.0, 20.0, 40.0 mg · kg⁻¹) did not increase ACT to a statistically significant level in comparison with the control group. However, pretreatment with AP5 (20.0 mg · kg⁻¹) was able to reverse the effect of NMDA on decreasing ACT in that ACT value was reversed from 159.6 ± 8.8 sec. (NMDA 10 mg · kg⁻¹) to 276.4 ± 11.2 s (pretreatment with AP5 20.0 mg · kg⁻¹) (Table II).

The combined effects of NMDA and L-NAME on hypoxic tolerance were shown in Table III. Pretreatment with AP-5 (20.0 mg · kg⁻¹) prevented the effect of NMDA (10.0 mg · kg⁻¹) on decreasing the hypoxic tolerance. In addition, this ACT decreasing effect caused by NMDA can also be prevented by pretreatment with an NOS inhibitor L-NAME (5.0 mg · kg⁻¹).

DISCUSSION

The major finding of the present study is that both NO and NMDA are involved in the mechanism of the acute hypobaric hypoxic tolerance. In addition, an

TABLE II. EFFECTS OF NMDA OR AP5 ADMINISTRATION ON THE ALTITUDE CONVULSION THRESHOLD (ACT) IN MICE.

Group	Altitude Convulsion Threshold (s)
I.	
Saline	226.6 ± 2.9
NMDA 2.5 mg · kg ⁻¹	228.1 ± 18.8
NMDA 10.0 mg · kg ⁻¹	165.6 ± 13.1*
NMDA 20.0 mg · kg ⁻¹	172.9 ± 7.6*
II.	
Saline	215.7 ± 6.0
AP5 10.0 mg · kg ⁻¹	214.7 ± 4.3
AP5 20.0 mg · kg ⁻¹	241.9 ± 23.8
AP5 40.0 mg · kg ⁻¹	242.6 ± 33.5
III.	
Saline	224.9 ± 9.1
NMDA 10.0 mg · kg ⁻¹	159.6 ± 8.8*
NMDA 10.0 mg · kg ⁻¹ + pretreatment with AP5 20.0 mg · kg ⁻¹	276.4 ± 11.2†

Data are expressed as the mean ± S.E.M. of 7 mice.

* Significantly different from the values in corresponding saline-treated mice, $p < 0.01$.

† Significantly different from the values in NMDA (10.0 mg · kg⁻¹)-treated mice, $p < 0.01$.

TABLE III. THE COMBINED EFFECTS OF NMDA, AP5 AND NMDA, L-NAME, AND NMDA ON THE TIME REQUIRED FOR INITIATING THE ALTITUDE CONVULSION IN MICE.

	Altitude Convulsion Threshold (s)
Saline	218.9 ± 16.4
NMDA 10.0 mg·kg ⁻¹	156.4 ± 16.4*
NMDA 10.0 mg·kg ⁻¹ + pretreatment with AP5 20.0 mg·kg ⁻¹	277.9 ± 28.1†
NMDA 10.0 mg·kg ⁻¹ + pretreatment with L-NAME 5.0 mg·kg ⁻¹	318.7 ± 37.7†

Data are expressed as the mean ± S.E.M. of 7 mice.

* Significantly different from the values in corresponding saline-treated mice, $p < 0.05$.

† Significantly different from the values in NMDA (10.0 mg·kg⁻¹)-treated mice, $p < 0.01$.

NOS inhibitor (L-NAME) blocks not only the neurotoxic effect induced by NO, but also that induced by NMDA in mouse altitude convulsion model. These findings suggest that nitric oxide may be involved in the sequence of events associated with development of altitude convulsion. It is well accepted that several neurotransmitters (e.g., dopamine, norepinephrine, serotonin, polyamine, GABA, and glutamate, etc.) are involved in the mechanisms of experimental convulsions (3,8), though few of these studies were investigated with altitude convulsion model. Our previous study (2) has confirmed that catecholaminergic mechanisms are involved in the regulation of hypoxic tolerance of altitude convulsion in mice. It is reasonable to expect that other neurotransmitters may also play a role in the pathophysiology of altitude convulsion. Recently, nitric oxide (NO) has been strongly suggested as a novel neuronal messenger and NO synthase (NOS) has a very wide distribution in the brain (7,16). In addition, NO is one of the key molecules that mediate the endothelium-dependent vasodilatation (15). The vasodilation is supposed to ameliorate the oxygen and energy demands in response to hypoxia (10). Faraci (4) reported that sustained dilatation of cerebral arterioles during seizures is dependent on production of NO which influences local vascular tone and this might be a major mechanism to couple local perfusion with neuronal activity. Because of its special characteristics as both neuronal messenger and vasodilator, NO is a strong candidate mediator in the induction and regulation of altitude convulsion.

The present study suggests the involvement of NO in the mechanisms of acute hypoxic altitude convulsion predominantly through its signalling or possible neurotoxic role in that administration of NO synthesis precursor L-arginine, which ameliorates the oxygen and energy supply in the brain under the severe acute hypoxic and hypobaric stress, did not increase the hypoxic tolerance in mice. On the contrary, it decreased the hypoxic tolerance and this deterioration caused by L-arginine could be prevented by pretreatment with NOS inhibitor, L-NAME. Abundant evidence implicates NO as a mediator of neurotoxicity, especially in response to glutamate through NMDA

receptor which augments NOS activity in brain slices (17). NMDA-induced neurotoxicity can be prevented by either NOS inhibitors or agents that indirectly block NO formation (17). Our findings that NMDA decreased the hypoxic tolerance and this decrease can be prevented by pretreatment with NOS inhibitor L-NAME provide further evidence for a role for nitric oxide in NMDA-receptor mediated neurotoxicity. These findings are consistent with a recent study in which blockade of the convulsive response to cocaine in mice by L-NAME was reported (9). The finding that L-NAME had no effect on the binding to the NMDA receptor complex also suggests that its protective effects against the convulsion induction is not mediated via a direct antagonism against the NMDA receptor (9). In addition, the recent findings that nitric oxide causes an increase in the release of norepinephrine (11) and dopamine (18) also provide further consistent explanations for the present and our previous studies (2) which found that either blockade of NO formation or depletion of catecholamines increased the hypoxic tolerance in mice.

Taken together, the present study suggests an important signalling role for nitric oxide and NMDA in the development of altitude convulsion and further supports the hypothesized relationship between NMDA-receptor mediated neurotoxicity and nitric oxide (17).

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